Simulation of Ground State Energy of H_2 using the Variational Quantum Eigensolver (VQE)

Abstract

Quantum chemistry aims to understand the behavior of molecules by solving the Schrödinger equation. However, as molecular systems become larger, the computational resources required to solve these problems exactly increase exponentially, making classical simulation extremely challenging. Quantum computing offers a promising alternative by using quantum systems to simulate other quantum systems naturally.

In this project, we implement the Variational Quantum Eigensolver (VQE), a hybrid quantum-classical algorithm, to estimate the ground state energy of the hydrogen molecule (H_2). The hydrogen molecule serves as an ideal benchmark because it is the simplest neutral molecule and its exact ground state energy is well known. Using Qiskit Nature, PySCF, Jordan-Wigner mapping, and a custom parameterized quantum circuit (ansatz), we calculate the molecular energy across a range of bond lengths and generate the corresponding potential energy curve. The project demonstrates how quantum computing can be applied to quantum chemistry problems and provides insights into the scalability, challenges, and future potential of VQE-based molecular simulations.

1. Introduction

Understanding the properties of molecules is one of the most important problems in physics, chemistry, and materials science. Predicting molecular behavior enables advancements in drug discovery, catalyst design, battery technology, and the development of new materials.

At the microscopic level, molecules obey the laws of quantum mechanics. To fully describe a molecule, we must solve the Schrödinger equation, which provides information about the molecule's wavefunction and energy levels. Unfortunately, the complexity of this problem increases exponentially with the number of particles involved. Even modern supercomputers struggle to accurately simulate relatively small molecular systems.

Quantum computers provide a promising solution because they themselves operate according to quantum mechanical principles. Instead of approximating a quantum system using classical resources, a quantum computer can naturally represent and manipulate quantum states. This makes quantum computing particularly attractive for molecular simulations.

In this project, we focus on the hydrogen molecule (H_2) and estimate its ground state energy using the Variational Quantum Eigensolver (VQE), one of the most widely studied quantum algorithms for near-term quantum devices.

2. Fundamentals of Quantum Computing

2.1 Classical Bits and Quantum Bits

A classical computer stores information using bits, which can exist in either state 0 or 1.

A quantum computer uses qubits. Unlike classical bits, a qubit can exist in a superposition of states:

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

where α and β are probability amplitudes satisfying:

$|\alpha|^2 + |\beta|^2 = 1$

This property allows quantum computers to represent many states simultaneously.

2.2 Superposition

Superposition is one of the most fundamental principles of quantum mechanics. A qubit in superposition simultaneously contains information about both the 0 and 1 states until a measurement is performed.

This property enables quantum computers to explore many possible solutions at the same time.

2.3 Entanglement

Entanglement is a uniquely quantum phenomenon where multiple qubits become correlated in such a way that the state of one qubit depends on the state of another.

Entanglement is essential for representing molecular interactions because electrons inside molecules are naturally correlated.

2.4 Quantum Gates

Quantum gates manipulate qubits similarly to how logic gates manipulate classical bits.

Important gates used in this project include:

RY Gate

The RY gate rotates a qubit around the Y-axis of the Bloch sphere.

It is used to create parameterized quantum states.

CNOT Gate

The Controlled-NOT (CNOT) gate entangles two qubits.

It is responsible for creating correlations between different parts of the quantum system.

3. Quantum Chemistry Background

3.1 The Schrödinger Equation

The behavior of a molecular system is described by the Schrödinger equation:

$H\psi = E\psi$

where:

H = Hamiltonian operator

ψ = Wavefunction

E = Energy

The solutions of this equation provide the allowed energy levels of the molecule.

3.2 Ground State Energy

The lowest energy solution of the Schrödinger equation is called the ground state energy.

The ground state is extremely important because it corresponds to the most stable configuration of a molecule.

Finding the ground state energy is one of the central objectives of computational chemistry.

3.3 Why Hydrogen (H_2)?

Hydrogen is chosen because:

- It is the simplest neutral molecule.
- It requires only a small number of qubits.
- Exact solutions are known.
- It serves as a standard benchmark for quantum algorithms.

Because of its simplicity, H_2 is often the first molecule used to test new quantum chemistry techniques.

4. Molecular Hamiltonian

The Hamiltonian represents the total energy of the molecular system.

The complete molecular Hamiltonian contains:

1. Electronic kinetic energy
2. Nuclear kinetic energy
3. Electron-electron repulsion
4. Electron-nucleus attraction
5. Nuclear-nuclear repulsion

Mathematically:

H = Te + Tn + Vee + Ven + Vnn

Solving this Hamiltonian exactly becomes computationally expensive as the system size grows.

4.1 Born-Oppenheimer Approximation

Nuclei are approximately 1800 times heavier than electrons.

Because of this large mass difference, nuclei move much more slowly than electrons.

The Born-Oppenheimer approximation assumes that nuclei remain fixed while solving the electronic problem.

This significantly simplifies molecular calculations.

5. Mapping Molecules to Qubits

Quantum computers operate on qubits rather than electrons.

Therefore, molecular operators must be transformed into a qubit representation.

This process is called fermion-to-qubit mapping.

5.1 Jordan-Wigner Transformation

In our implementation, we use the Jordan-Wigner mapping.

This transformation converts fermionic creation and annihilation operators into combinations of Pauli operators.

After mapping, the molecular Hamiltonian becomes a weighted sum of Pauli strings such as:

I

Z_0

Z_1

X_0X_1

Y_0Y_1

These operators can be executed directly on quantum hardware.

6. Variational Quantum Eigensolver (VQE)

The Variational Quantum Eigensolver is a hybrid quantum-classical algorithm used to estimate ground state energies.

Instead of solving the Schrödinger equation directly, VQE searches for a quantum state with minimum energy.

The algorithm combines the strengths of both quantum and classical computation.

6.1 Variational Principle

The foundation of VQE is the variational principle.

For any trial quantum state:

$E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle \geq E_0$

where:

E_0 = True ground state energy

This principle guarantees that minimizing $E(\theta)$ moves us closer to the true ground state.

6.2 Hybrid Quantum-Classical Optimization

Quantum Computer:

- Prepares trial states
- Measures energy

Classical Computer:

- Updates parameters

• Searches for lower energy configurations

This process continues until convergence.

7. Ansatz Design

The ansatz is a parameterized quantum circuit used to prepare trial quantum states.

The quality of the ansatz strongly influences the performance of VQE.

7.1 Our Ansatz

We implemented a hardware-efficient ansatz using:

• Two layers of RY rotations

• One entanglement layer using CNOT gates

The circuit contains eight trainable parameters.

7.2 Layer-by-Layer Explanation

Layer 1: RY Rotations

Purpose:

Create superposition and explore different quantum states.

Entanglement Layer

Purpose:

Create correlations between qubits and model electron interactions.

Implemented using:

CNOT(0,1)

CNOT(1,2)

CNOT(2,3)

Layer 2: RY Rotations

Purpose:

Refine and fine-tune the quantum state for improved accuracy.

7.3 Why This Ansatz?

This ansatz was chosen because:

• Simple implementation

• Low circuit depth

• Suitable for near-term quantum hardware

• Sufficient expressibility for H_2

8. Energy Evaluation

For each set of parameters θ:

1. Construct quantum circuit
2. Generate quantum state
3. Calculate expectation value

The energy is computed as:

$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$

This value serves as the objective function for optimization.

9. Classical Optimization

To minimize the energy, we use the COBYLA optimizer.

COBYLA stands for Constrained Optimization BY Linear Approximation.

Why COBYLA?

• Gradient-free

• Stable

• Widely used in VQE

• Suitable for noisy quantum environments

Warm Start Strategy

Instead of starting optimization from random parameters each time, we reuse the optimized parameters from the previous bond length.

Benefits:

- Faster convergence
- Reduced optimization time
- Improved stability

Since neighboring bond lengths have similar optimal states, warm-starting significantly improves efficiency.

10. Implementation Workflow

The complete workflow is:

Step 1: Define bond length.

Step 2: Construct molecular Hamiltonian using PySCF.

Step 3: Apply Jordan-Wigner transformation.

Step 4: Create parameterized ansatz.

Step 5: Generate quantum state.

Step 6: Compute expectation value.

Step 7: Minimize energy using COBYLA.

Step 8: Store optimized energy.

Step 9: Repeat for all bond lengths.

Step 10: Generate energy curve and convergence plots.

11. Results

11.1 Energy vs Bond Length

The energy of the hydrogen molecule was computed for bond lengths ranging from 0.5 Å to 3.5 Å.

The resulting energy curve reveals how molecular stability changes with interatomic distance.

The minimum point on this curve corresponds to the equilibrium bond length.

This represents the most stable configuration of the molecule.

11.2 Convergence Plot

The convergence plot shows how the energy changes during optimization.

Observations:

• Energy decreases steadily.

• Optimizer successfully finds lower-energy states.

• Convergence occurs within a finite number of iterations.

This demonstrates the effectiveness of the VQE algorithm.

12. Physical Interpretation

The energy curve provides direct physical insight into molecular bonding.

Small Bond Lengths

When atoms are very close together:

• Nuclear repulsion dominates.

• Energy becomes very large.

• Molecule becomes unstable.

Equilibrium Bond Length

At the minimum energy point:

• Attractive and repulsive forces balance.

• Molecule is most stable.

• Chemical bond is formed.

Large Bond Lengths

When atoms move far apart:

• Bond weakens.

• Energy approaches that of two isolated hydrogen atoms.

• Molecule dissociates.

13. Challenges Encountered

Several challenges were encountered during implementation.

Hamiltonian Construction

Understanding the transformation from molecular operators to qubit operators required significant study of quantum chemistry concepts.

Ansatz Selection

A deeper ansatz improves accuracy but increases optimization complexity.

Balancing expressibility and simplicity was necessary.

Optimization Difficulties

Optimization landscapes may contain local minima.

Choosing an appropriate optimizer was important for convergence.

Computational Cost

VQE requires many energy evaluations.

The optimization process can become expensive for larger systems.

14. Scalability Analysis

Although H_2 requires only four qubits, larger molecules present significant challenges.

More Qubits

Larger molecules require:

• More orbitals

• More qubits

• Larger Hamiltonians

Measurement Overhead

The number of Pauli terms grows rapidly with system size.

More measurements become necessary.

Optimization Complexity

Larger ansätze introduce:

• More parameters

• Harder optimization

• Longer execution times

Barren Plateaus

As circuits become deeper, gradients may vanish.

This makes optimization increasingly difficult.

Noise

Current quantum hardware suffers from:

• Gate errors

• Decoherence

• Measurement noise

These effects reduce accuracy.

15. Future Scope

Several directions exist for extending this work.

1. Simulate larger molecules such as LiH and BeH_2.
2. Implement chemically inspired ansätze such as UCCSD.

3. Run experiments on real IBM Quantum hardware.
4. Apply error mitigation techniques.
5. Explore adaptive VQE methods.
6. Investigate fault-tolerant quantum computing approaches.

As quantum hardware improves, these techniques may enable simulations that are impossible on classical computers.

16. Conclusion

In this project, we successfully implemented the Variational Quantum Eigensolver to estimate the ground state energy of the hydrogen molecule.

The project demonstrated the complete workflow of quantum chemistry simulation, including Hamiltonian construction, qubit mapping, ansatz design, energy evaluation, and classical optimization.

The generated energy curve accurately captured the physical behavior of molecular bonding and identified the equilibrium bond length of H_2.

Although current quantum hardware remains limited, VQE represents one of the most promising approaches for near-term quantum chemistry applications. The techniques explored in this project provide a foundation for studying larger and more complex molecular systems in the future.

This work highlights the potential of hybrid quantum-classical algorithms and illustrates how quantum computing may transform computational chemistry in the coming decades.